

Students, in groups of two to three people, should identify a suitable scenario in which they test the behavioural sensitivity of an intervention, implemented in an agent-based setting. The complete case, supported with relevant literature, should be written up and submitted in the form of a peer-reviewed journal/conference article of no more than 6 pages, all inclusive. Each group is required to submit three deliverables:

1. The portable document format (PDF) version of the final manuscript, preferably prepared using the LATEX template provided on Toledo.
2. A reproducible and self-contained version of the model. This can be either a compressed folder containing the entire (AnyLogic) model, or a URL to a KU Leuven GitLab repository that can be cloned. The latter is preferred.
3. A self-contained presentation of the overall project. This could be narrated slides or a knowledge clip. You can pick the medium and style of the deliverable.

- The following rubric is used to assess the group's submission of the course project along eight criteria, each evaluated on a 0–1–2–3 basis. Remember that while the group's submission is marked, each student's final mark will be affected by the peer assessment.

1. Relevance and suitability. This criterion assesses whether the problem you chose is meaningful, well-scoped, and appropriate for AgBM.
 0. Problem unclear, trivial, or wildly over-complex; AgBM not justified.
 1. Problem is described but not well-motivated; AgBM seems unnecessary or poorly justified.
 2. Good problem selection with clear reasoning for using AgBM; scope is appropriate.
 3. Excellent choice. Non-trivial, interesting, and well-targeted; very strong justification for AgBM.
2. Literature and conceptual framing. Assesses how well you connect your project to high-quality academic sources. This is not about quantity but about quality and the consistency of the citations and the bibliography.
 0. No relevant literature cited; lacks theoretical grounding.
 1. Minimal or superficial use of literature; weak integration.
 2. Solid use of peer-reviewed AgBM literature that informs your design.
 3. Elegant, concise synthesis of relevant work that clearly strengthens your model's foundations.
3. Problem reduction and experimental design. This evaluates your ability to simplify real-world systems to their essential mechanisms and to design meaningful experiments.
 0. No clear abstraction; key elements missing or irrelevant ones included.
 1. Simplification attempted but flawed or poorly justified.
 2. Good reduction to essential components with a clear, purposeful experimental plan.
 3. Clean, elegant abstraction that aligns fully with your goals; experiments are thoughtfully de-

signed to reveal insights.

4. Model architecture and implementation, focusing on how clearly and rigorously you specify your agents, interactions, environment, and update rules.
 0. Architecture is unclear, incorrect, or missing.
 1. Model exists but is poorly explained or hard to understand.
 2. Clear and well-structured architecture; diagrams and descriptions support understanding.
 3. Exceptionally clear, elegant, and well-documented architecture; highly professional presentation.
5. Experiments, results, and interpretation. Here it is about how rigorously you run and interpret your experiments.
 0. No experiments or invalid results.
 1. Results exist but lack rigour (few runs, unclear methods, weak interpretation).
 2. Solid experimental design with clear results and reasonable interpretation.
 3. High-quality experimental work: systematic, well-visualised, thoughtful interpretation, and

awareness of limitations.

6. Report structure, graphics, and presentation quality. Evaluates whether your written report is organised, readable, and professionally presented.
 0. Disorganised; missing sections; unreadable graphics.
 1. Structure present but inconsistent; figures unclear or poorly integrated.
 2. Well-organised, clear report with good figures and academic structure.
 3. Polished, conference-style document; excellent figures fully integrated into the narrative.
7. Writing quality and narrative coherence. Assesses the clarity and elegance of your writing.
 0. Writing unclear or difficult to follow.
 1. Understandable but awkward or imprecise.
 2. Clear, coherent, and academically appropriate (without the feeling that it was massaged by AI).
 3. Exceptionally clear, engaging, and elegant writing that enhances understanding.
8. Presentation/communication in an alternative medium. Evaluates your ability to communicate your project to a broader, possibly non-technical audience using your chosen medium (e.g., short video, narrated slides, knowledge clip).
 0. No presentation or unclear/incoherent content.
 1. Presentation is overly technical, hard to follow, or lacks structure/creativity.

2. Clear and engaging explanation with a logical flow and appropriate use of the chosen medium.
3. Highly compelling, creative, and professional storytelling; excellent audience awareness; polished

use of the medium.

For the peer assessment, each student evaluates both their own contribution and that of each of the other members along the following criteria:

1. Attendance. Are the group members actively availing themselves to collaborate (in person or online), and then adhering to those commitments?
 - o nline), and then adhering to those commitments?
 2. Participation. Are the members (pro)actively participating in the group, or are there some that you have to pull everything out of them, or, on the other end of the spectrum, are overbearing?
 3. Autonomy. When work is distributed, are members doing what they've been assigned to do? And are they only doing what they've been told, or is there initiative being taken?
 4. Technical valued-add. What are the members' contributions regarding the agent-based modelling? This criterion specifically speaks to the science of content ; the technical aspects of modelling and solving the problem.
 5. Packaging value-add. What are the members' contributions regarding the presentation of the work? This criterion specifically speaks to the science of packaging ; the report writing, and the preparation of the alternative medium deliverable.
- For each criterion, you have 30 marks to distribute among the group members, and only integer values are allowed. For example, in a group of 3 people, if you want to give 12/30 to one person for their autonomy and initiative, the remaining 18 marks (for that criterion) must be distributed among the remaining members. Peer assessments are anonymous and are submitted independently of the group's submission. Each group will be sent a prepopulated sheet to complete. The individually completed form (electronically completed or a hard copy scanned) is then submitted via a separate individual assignment created for this in Toledo.

Grader rubric: Multi-agent systems (H02H4a)

Group:

1. Relevance and suitability
2. Literature and conceptual framing
3. Problem reduction and experimental design
4. Model architecture and implementation
5. Experiments, results, and interpretation
6. Report structure, graphics, and presentation quality
7. Writing quality and narrative coherence
8. Alternative medium

Competent

Not attempted or fundamentally incorrect

0	1	2	3
0	1	2	3
0	1	2	3
0	1	2	3
0	1	2	3
0	1	2	3
0	1	2	3
0	1	2	3

Major Meeting issues Master's Below expectations

Master's expectation

Room for

Exemplary Exceeding Master's

improvement expectations

Comments/notes: